Pembrolizumab plus trastuzumab and chemotherapy for HER2-positive gastric or gastro-oesophageal junction adenocarcinoma: interim analyses from the phase 3 KEYNOTE-811 randomised placebo-controlled trial

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Summary

Background Evidence for the efficacy of combined PD-1 and HER2 blockade with chemotherapy on progression-free and overall survival in HER2-positive gastro-oesophageal cancer is scarce. The first interim analysis of the randomised, phase 3 KEYNOTE-811 study showed a superior objective response with pembrolizumab compared with placebo when added to trastuzumab plus fluoropyrimidine and platinum-based chemotherapy. Here, we report results from protocol-specified subsequent interim analyses of KEYNOTE-811.

Methods The randomised, phase 3 KEYNOTE-811 trial involved 168 medical centres in 20 countries worldwide. Patients aged 18 years or older with locally advanced or metastatic HER2-positive gastro-oesophageal junction adenocarcinoma, without previous first-line treatment, were randomly assigned (1:1) by an integrated interactive voice-response and web-response system to intravenous pembrolizumab 200 mg or placebo, both to be combined with standard chemotherapy (fluoropyrimidine and platinum-based therapy) plus trastuzumab every 3 weeks for up to 35 cycles or until disease progression, unacceptable toxic effects, or investigator or participant-initiated withdrawal. Randomisation used a block size of four and was stratified by region, PD-L1 status, and chemotherapy. Dual primary endpoints were progression-free and overall survival, analysed by intention to treat. Safety was assessed in all randomly assigned patients who received at least one dose of study treatment according to the treatment received. KEYNOTE-811 is registered with ClinicalTrials.gov (NCT03615326) and is active but not recruiting.

Findings Between Oct 5, 2018, and Aug 6, 2021, 698 patients were assigned to pembrolizumab (n=350) or placebo (n=348). 564 (81%) were male and 134 (19%) were female. At the third interim analysis, 286 (82%) of 350 patients in the pembrolizumab group and 304 (88%) of 346 in the placebo group who received treatment had discontinued treatment, mostly due to disease progression. At the second interim analysis (median follow-up 28·3 months [IQR 19·4–34·3] in the pembrolizumab group and 28·5 months [20·1–34·3] in the placebo group), median progression-free survival was 10·0 months (95% CI 8·6–11·7) in the pembrolizumab group versus 8·1 months (7·0–8·5) in the placebo group (hazard ratio [HR] 0·72, 95% CI 0·60–0·87; p=0·0002). Median overall survival was 20·0 months (17·8–23·2) versus 16·9 months (15·0–19·8; HR 0·87 [0·72–1·06]; p=0·084). At the third interim analysis (median follow-up 38·4 months [IQR 29·5–44·4] in the pembrolizumab group and 38·6 months [30·2–44·4] in the placebo group), median progression-free survival was 10·0 months (8·6–12·2) versus 8·1 months (7·1–8·6; HR 0·73 [0·61–0·87]), and median overall survival was 20·0 months (17·8–22·1) versus 16·8 months (15·0–18·7; HR 0·84 [0·70–1·01]), but did not meet prespecified criteria for significance and will continue to final analysis. Grade 3 or worse treatment-related adverse events occurred in 204 (58%) of 350 patients in the pembrolizumab group versus 176 (51%) of 346 patients in the placebo group. Treatment-related adverse events that led to death occurred in four (1%) patients in the pembrolizumab group and three (1%) in the placebo group. The most common treatment-related adverse events of any grade were diarrhoea (165 [47%] in the pembrolizumab group *vs* 145 [42%] in the placebo group), nausea (154 [44%] *vs* 152 [44%]), and anaemia (109 [31%] *vs* 113 [33%]).

Interpretation Compared with placebo, pembrolizumab significantly improved progression-free survival when combined with first-line trastuzumab and chemotherapy for metastatic HER2-positive gastro-oesophageal cancer, specifically in patients with tumours with a PD-L1 combined positive score of 1 or more. Overall survival follow-up is ongoing and will be reported at the final analysis.

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See Online for appendix

Research in context

Evidence before this study

For more than a decade combination therapy with the anti-HER2 antibody trastuzumab and chemotherapy has been standard first-line therapy for patients with HER2-positive gastric or gastro-oesophageal junction adenocarcinomas. Although checkpoint inhibitors in combination with chemotherapy are an effective treatment option for patients with HER2-negative gastric or gastro-oesophageal junction cancer, data on their effect in combination with standard therapy for treatment of HER2-postive gastric or gastro-oesophageal junction cancer are scarce. On July 13, 2023, we conducted a PubMed search for English language papers using the terms "PD-1" OR "PD-L1" OR "pembrolizumab" OR "nivolumab" OR "atezolizumab" OR "durvalumab" OR "avelumab" AND "HER2-positive gastric cancer" without any publication date restrictions. We identified two studies evaluating first-line therapy with an anti-PD-1 and anti-PD-L1 plus anti-HER2 antibody with or without chemotherapy in patients with HER2-positive advanced gastric or gastro-oesophageal junction cancer: the phase 3 randomised KEYNOTE-811 study of pembrolizumab plus trastuzumab and chemotherapy, and the phase 2 open-label INTEGA study of ipilimumab or FOLFOX (leucovorin, fluorouracil, and oxaliplatin) in combination with nivolumab and trastuzumab. Initial data from KEYNOTE-811 showed that the combination of pembrolizumab plus trastuzumab and chemotherapy significantly improved the objective response rate compared with trastuzumab and chemotherapy in patients with unresectable or metastatic HER2-positive gastric or gastro-oesophageal junction cancer and resulted in the US Food and Drug Administration granting accelerated approval of pembrolizumab in combination with trastuzumab and fluoropyrimidine-containing and platinum-containing chemotherapy for first-line treatment of locally advanced unresectable or metastatic HER2-positive gastric or gastro-oesophageal junction cancer.

Added value of this study

Data from this interim analysis of KEYNOTE-811 add to the little evidence evaluating combined blockade with anti-PD-1 and anti-HER2 therapy plus chemotherapy on progression-free and overall survival in patients with HER2-positive gastric or gastro-oesophageal junction cancer. The study met the prespecified boundary for detecting superiority in progression-free survival with pembrolizumab plus trastuzumab and chemotherapy compared with trastuzumab and chemotherapy at the second interim analysis, with a manageable safety profile. The effect was enhanced in patients with tumours with a PD-L1 combined positive score (CPS) of 1 or more with little to no benefit in patients with tumours with a PD-L1 CPS of less than 1. The study continues to the final analysis of overall survival.

Implications of all the available evidence

The addition of pembrolizumab to first-line trastuzumab and chemotherapy led to meaningfully longer progression-free survival and improvement in tumour response, particularly in the subgroup of patients with HER2-positive and PD-L1-positive tumours. These data support the use of first-line anti-PD-1 therapy in combination with trastuzumab in first-line HER2-positive gastric or gastro-oesophageal junction cancer and are the first to show a progression-free survival improvement over standard of care in this patient population. The effect of this regimen on progression-free survival suggests that it could be considered as a first-line therapy option for patients with metastatic gastro-oesophageal cancer with co-occurring HER2 and PD-L1 overexpression; however, the overall survival benefit of this regimen is yet to be determined.

Introduction

Approximately one quarter of gastric and gastro-oesophageal junction adenocarcinomas are characterised by aberrant activation of human epidermal growth factor receptor (HER) tyrosine kinases.[1–4] In patients whose cancers harbour *HER2* (also known as *ERBB2*) amplification or HER2 protein overexpression, the combination of the anti-HER2 antibody trastuzumab with cytotoxic chemotherapy (a fluoropyrimidine and platinum) is a recommended first-line therapy.[5] The combination of tumour-targeting antibodies with PD-1 inhibition has been shown in preclinical studies to improve immune infiltration, T-cell response, and help overcome tolerogenic dendritic cells,[6,7] translating to meaningful responses with pembrolizumab, trastuzumab, and chemotherapy in patients with HER2-positive gastro-oesophageal cancer.[8,9] Anti-PD-1 antibodies have been shown to improve survival relative to first-line chemotherapy in patients with HER2-negative, metastatic gastro-oesophageal cancer, with enhanced benefit seen in patients with tumours with a PD-L1 combined positive score (CPS) of 1 or more.[10,11]

The randomised, phase 3 KEYNOTE-811 study (NCT03615326) evaluated the efficacy and safety of the addition of either pembrolizumab or placebo to trastuzumab plus fluoropyrimidine and platinum-based chemotherapy (fluorouracil plus cisplatin or capecitabine plus oxaliplatin) in patients with untreated HER2-positive metastatic gastric or gastro-oesophageal junction cancer.[12] The study met the prespecified statistical parameters for success at the first interim analysis, showing a superior objective response with pembrolizumab compared with placebo.[12] Here, we report results from protocol-specified subsequent interim analyses of KEYNOTE-811.

Methods

Study design and participants

KEYNOTE-811 is a randomised, double-blind, placebo-controlled, phase 3 trial done at 168 medical centres in

20 countries worldwide. Study eligibility criteria for KEYNOTE-811 have been previously published.[12] Eligible patients were aged 18 years or older, with previously untreated, unresectable or metastatic, pathologically confirmed gastric or gastro-oesophageal junction adenocarcinoma that was HER2-positive (either with an immunohistochemistry test score of 3+ [IHC 3+] or IHC 2+ with positive in-situ hybridisation or fluorescence in-situ hybridisation) as assessed by central review. Patients also had to have measurable disease per Response Evaluation Criteria in Solid Tumours (RECIST) version 1.1 (v1.1),[13] an Eastern Cooperative Oncology Group performance status of 0 or 1,[14] life expectancy of greater than 6 months, adequate organ function, and provided a tumour sample adequate for assessment of PD-L1 and microsatellite instability status. Full eligibility criteria are listed in the study protocol (appendix pp 73–75). The protocol and all amendments were approved by the relevant institutional review board or independent ethics committee at each study centre. The study was conducted in accordance with the protocol, Good Clinical Practice guidelines, and the Declaration of Helsinki. All patients provided written informed consent.

Randomisation and masking

Patients were randomly assigned centrally in a 1:1 ratio with a block size of four to receive either pembrolizumab or placebo by an integrated interactive voice-response and web-response system (Almac Clinical Technologies, Souderton, PA, USA). Randomisation was stratified by geographical region (Australia, Europe, Israel, and North America *vs* Asia *vs* rest of world), PD-L1 CPS (≥1 *vs* <1), and investigator's choice of chemotherapy (fluorouracil plus cisplatin *vs* capecitabine plus oxaliplatin). Investigators and site personnel were responsible for enrolling participants and assigning them to the trial groups via the interactive voice-response system; these individuals were also involved in other aspects of the trial. All patients and investigators involved in study treatment administration or clinical evaluation of the patients were masked to the group assignments in a double-blinded manner as described in the protocol (appendix p 84). Final unmasking of the database will occur upon completion of medical and scientific review, identification of protocol deviations, and when data are final and complete.

Procedures

Patients received either 200 mg pembrolizumab or placebo (normal saline or dextrose) intravenously once every 3 weeks. In addition, all patients received trastuzumab at a dose of 6 mg/kg intravenously once every 3 weeks following an initial loading dose of 8 mg/kg, and either fluorouracil (800 mg/m² administered intravenously on days 1–5 of each 3-week cycle) and cisplatin (80 mg/m² administered intravenously once every 3 weeks) or capecitabine (1000 mg/m² administered orally twice daily on days 1–14 of each 3-week cycle) and oxaliplatin (130 mg/m² administered intravenously once every 3 weeks). Treatment continued for up to 35 cycles (about 2 years) or until disease progression, unacceptable toxic effects, investigator decision to discontinue treatment, or patient withdrawal of consent. Imaging was performed at week 6 and every 6 weeks thereafter. Response was assessed according to RECIST v1.1, by blinded independent central review for determination of study endpoints, and according to immune RECIST (iRECIST)[15] by the investigator to inform treatment decisions. PD-L1 expression was assessed by the PD-L1 IHC 22C3 pharmDx assay (Agilent Technologies, Carpinteria, CA, USA) according to the CPS, with a minimum of 100 viable tumour cells.[16] Microsatellite instability status was determined centrally by PCR using the MSI Analysis System, version 1.2 (Promega, Madison, WI, USA), and characterised as high if two or more tumour microsatellite loci were changed compared with normal control samples. Data on adverse events were collected regularly during study treatment and for up to 30 days thereafter (up to 90 days for serious adverse events) and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.03. Following treatment discontinuation, patients were assessed for survival every 12 weeks. Demographic data were self-reported and collected by the investigator; sex was self-reported as either male or female.

Outcomes

The dual primary endpoints were progression-free survival per RECIST v1.1, by blinded independent central review (from randomisation to first disease progression or death) and overall survival (from randomisation to death from any cause) in the intention-to-treat (ITT) population. Secondary endpoints included objective response (proportion of patients with complete or partial response) and duration of response (time from first complete or partial response until first disease progression) per RECIST v1.1, by blinded independent central review and safety. Exploratory endpoints included health-related quality of life as evaluated by the European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30, EORTC QLQ-STO22, and EQ-5D-5L scales, identification of molecular, genetic, and genomic correlates of treatment, and progression-free survival and overall response according to iRECIST. Complete study and treatment details are in the protocol (appendix pp 62–82).

Statistical analysis

The statistical analysis plan specified three interim analyses and a final analysis. An external data safety monitoring committee reviewed interim study results to ensure patient safety and to assess efficacy at prespecified interim analyses. A sample size of 692 patients was planned. The assumptions that contributed to the planned

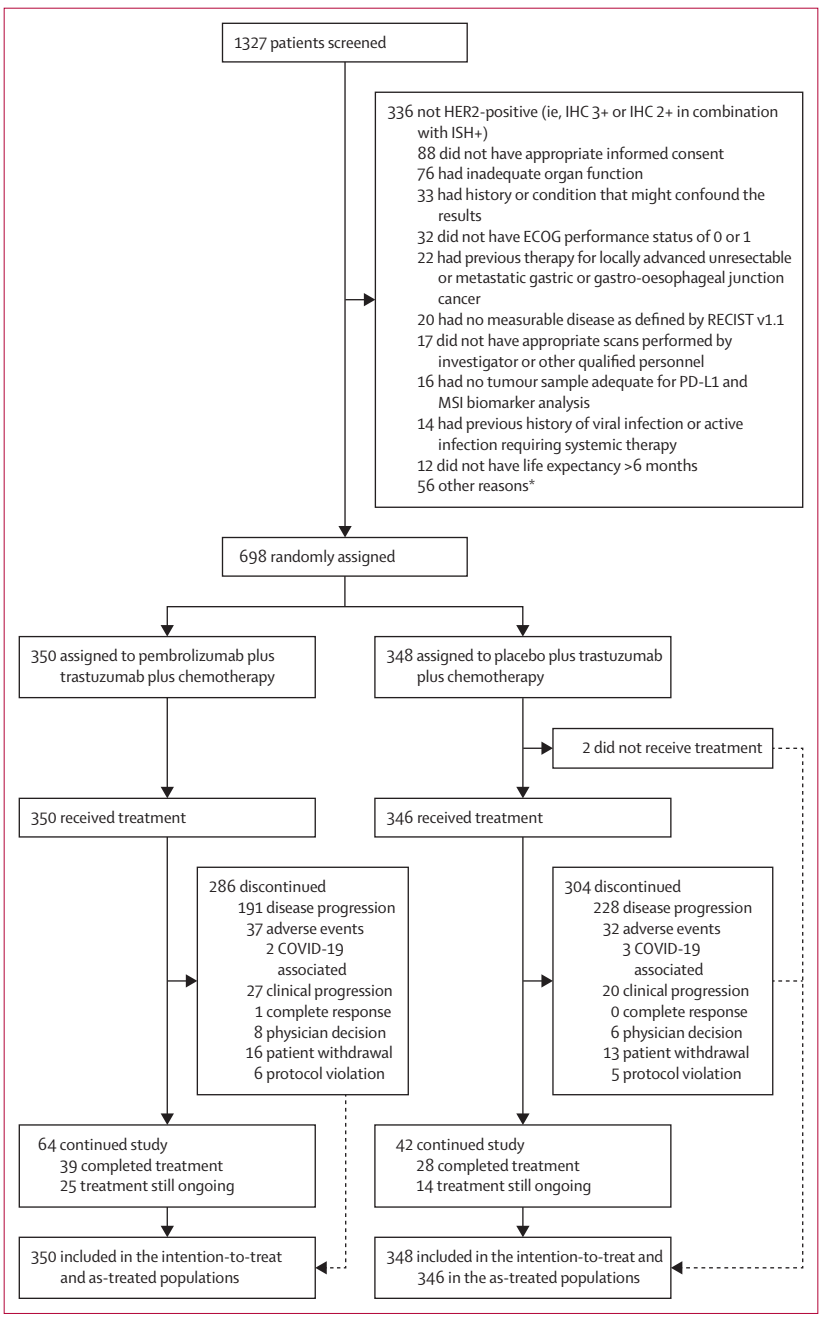

Figure 1: **Study disposition at third interim analysis**
Patients were randomly assigned to pembrolizumab or placebo plus trastuzumab and investigator choice of chemotherapy. Please note that a patient might appear more than once in the breakdown of reasons for ineligibility. ECOG=Eastern Cooperative Oncology Group. IHC=immunohistochemistry test score. ISH=in-situ hybridisation. MSI=microsatellite instability. RECIST v1.1=Response Evaluation Criteria in Solid Tumours version 1.1. *Other reasons included no adequate method of contraception or were pregnant or breastfeeding (n=8); accumulation of fluid requiring draining (n=7); symptomatic heart failure or clinically significant cardiac disease (n=6); active autoimmune disease requiring systemic treatment or diagnosis of immunodeficiency (n=6); a known additional malignancy (n=5); had received previous checkpoint inhibitor therapy or used an investigational agent or device, or had live vaccine within 4 weeks of first dose (n=5); major surgery within 28 days of randomisation (n=4); known CNS active metastases (n=4); a history of non-infectious pneumonitis or active tuberculosis (n=4); poorly controlled diarrhoea (n=2); grade 1 or worse peripheral neuropathy (n=2); radiotherapy within 14 days of randomisation (n=1); grade 3 or worse hypersensitivity to any pembrolizumab excipients (n=1); and allogeneic organ transplant (n=1).

sample size included the planned enrolment period, and assumptions of median progression-free survival and overall survival, which are detailed in the appendix (pp 136–52). The first interim analysis (hypothesis testing of objective response rate) occurred when the first 260 patients enrolled had at least 8·5 months of follow-up, the results of which have been published.[12] The second interim analysis was the first assessment of progression-free and overall survival and occurred at least 9 months after the last patient was randomly assigned. The third interim analysis (final analysis of progression-free and interim analysis of overall survival) occurred at least 18 months after the last patient was randomly assigned, and the final analysis of overall survival is planned for when at least 28 months have passed since the last patient was randomly assigned and 551 deaths have occurred. The independent data monitoring committee confirmed that the study met the prespecified endpoint of objective response rate at first interim analysis and met the prespecified endpoint of progression-free survival at the second interim analysis. As such, there was no formal hypothesis testing of and therefore, no p values provided for, objective response at the second or third interim analyses or of progression-free survival at the third interim analysis. All data reported here are from the second (data cutoff date of May 25, 2022) and third interim analyses (data cutoff date of March 29, 2023).

Efficacy was assessed in the ITT population of all randomly assigned patients according to their allocated treatment. Safety was assessed in the as-treated population (all randomly assigned patients who received at least one dose of study treatment according to the treatment received). The non-parametric Kaplan-Meier method was used to estimate progression-free survival and overall survival per prespecified censoring rules. For the primary progression-free survival analysis, if the events of progression or death were immediately after more than one missed disease assessment or after starting new anticancer therapy, the data were censored at the last assessment before the missing visit or new anticancer therapy. For overall survival, patients without documented death at the time of analysis were censored at the date of last contact. Duration of response was summarised descriptively using Kaplan-Meier median and quartiles. Patients without assessments were considered to have not responded and were included in the denominator. Between-group differences in progression-free and overall survival were assessed using the stratified log-rank test. The consistency of the progression-free survival treatment effect in protocol-specified subgroups was assessed descriptively by means of hazard ratios (HRs) and associated 95% CIs calculated using the non-stratified Cox proportional hazards model with Efron's method of tie handling. The proportional hazard assumption was assessed via examining scaled Schoenfeld residuals. A stratified Cox proportional hazards model with Efron's method of tie handling was used to estimate HRs and

	Pembrolizumab group (N=350)	Placebo group (N=348)
Age, years	62 (54–69)	63 (55–70)
≥65 years	145 (41%)	156 (45%)
Sex		
Female	66 (19%)	68 (20%)
Male	284 (81%)	280 (80%)
Race		
American Indian or Native Alaskan	5 (1%)	6 (2%)
Asian	119 (34%)	121 (35%)
Black or African American	2 (1%)	2 (1%)
Multiple	6 (2%)	5 (1%)
White	218 (62%)	212 (61%)
Missing	0	2 (1%)
Geographical region		
Australia, Europe, Israel, North America	113 (32%)	111 (32%)
Asia	118 (34%)	119 (34%)
Rest of world	119 (34%)	118 (34%)
ECOG performance status		
0	146 (42%)	146 (42%)
1	204 (58%)	202 (58%)
Primary location at diagnosis		
Gastro-oesophageal junction	110 (31%)	122 (35%)
Stomach	240 (69%)	226 (65%)
Metastatic sites		
0–2	179 (51%)	198 (57%)
≥3	171 (49%)	150 (43%)
Histological subtype		
Diffuse	68 (19%)	51 (15%)
Intestinal	198 (57%)	188 (54%)
Indeterminate	83 (24%)	109 (31%)
Unknown	1 (<1%)	0
Primary gastrectomy or oesophagectomy		
Yes	51 (15%)	63 (18%)
No	299 (85%)	285 (82%)

(Table 1 continues in next column)

	Pembrolizumab group (N=350)	Placebo group (N=348)
(Continued from previous column)		
PD-L1 status		
CPS ≥1	298 (85%)	296 (85%)
CPS <1	52 (15%)	52 (15%)
HER2 status, n (%)		
IHC 1+	1 (<1%)	1 (<1%)
IHC 2+ ISH equivocal	0	1 (<1%)
IHC 2+ ISH negative*	1 (<1%)	1 (<1%)
IHC 2+ ISH positive	62 (18%)	84 (24%)
IHC 3+	286 (82%)	261 (75%)
Microsatellite instability status		
High	6 (2%)	2 (1%)
Non-high	326 (93%)	329 (95%)
Unknown	18 (5%)	17 (5%)
Tumour burden		
<Median	162 (46%)	166 (48%)
≥Median	171 (49%)	170 (49%)
Missing	17 (5%)	12 (3%)
Chemotherapy regimen		
Capecitabine and oxaliplatin	297 (85%)	299 (86%)
Fluorouracil and cisplatin	53 (15%)	49 (14%)

Data are median (IQR) or n (%). Data are shown for the intention-to-treat population at the third interim analysis. Percentages might not total 100 because of rounding. CPS=combined positive score. ECOG=Eastern Cooperative Oncology Group. IHC=immunohistochemistry test score. ISH=in-situ hybridisation. *Some patients with IHC 1+ and IHC 2+ ISH-negative tumours were included per investigator assessment.

Table 1: **Demographics and patient characteristics at baseline**

associated 95% CIs for the analyses of the whole study population. The randomisation stratification factors were applied to both the stratified log-rank test and Cox model. Patients who have curative surgical resection during the study will be followed up for progression-free survival events after surgery until local recurrence, distant metastasis, or death for the primary progression-free survival analysis. An extension of the graphical method of Mauer and Bretz was used to control the overall type 1 error at a one-sided α-level of 0·025, with 0·002 originally allocated to objective response rate, 0·003 allocated to progression-free survival, and 0·02 to overall survival. Cumulative α spending at interim analyses was based on the Lan-DeMets O'Brien-Fleming spending function for reallocation of α. Statistical boundaries at each interim analysis were updated using a Lan-DeMets O'Brien-Fleming spending fraction on the basis of the information and α-level of the endpoint at each interim analysis. The study was not powered to test hypotheses in the patient subgroups, and these analyses were not controlled for α spending. The study is considered successful if pembrolizumab is superior to placebo for either primary endpoint.

Clinically important protocol deviations occurred in four patients in the pembrolizumab group (two patients received trastuzumab that had been improperly stored and deemed unacceptable for use and two patients were dispensed additional doses of capecitabine), and in one patient in the placebo group (ineligible due to clinically significant cardiac disease). A table of protocol deviations is included in the appendix (p 12). Missing data for any patients were managed via prespecified data handling and analysis guidelines.

All statistical analyses were performed using SAS software, version 9.4. The full statistical analysis plan is in the protocol in the appendix (pp 136–50). KEYNOTE-811 is registered with ClinicalTrials.gov, number NCT03615326.

Role of the funding source

The funder of this study participated in the study design, data collection, data analysis, data interpretation, and the

writing of this report and maintained the study database. Assistance in the preparation of the manuscript was provided by a medical writer employed by the funder.

Results

From Oct 5, 2018, to Aug 6, 2021, 1327 patients were screened and 698 randomly assigned to pembrolizumab (pembrolizumab group [n=350]) or placebo (placebo group [n=348]) in combination with trastuzumab and chemotherapy (figure 1). Patient demographics and baseline disease characteristics were generally well balanced between groups. Patients had a median age of 63 years (IQR 54–70), 681 (98%) had metastatic disease (340 [97%] in the pembrolizumab group and 341 [98%] in the placebo group), and 17 (2%) had locally advanced disease (10 [3%] in the pembrolizumab group and 7 [2%] in the placebo group). 466 (67%) of all 698 patients had primary gastric cancer, 547 (78%) had tumours with HER2 overexpression of IHC 3+, and 594 (85%) had tumours with a PD-L1 CPS of 1 or more. Capecitabine plus oxaliplatin was the preferred chemotherapy combination selected by the investigators for 596 (85%) patients (table 1).

At the second interim analysis, the median time from randomisation to data cutoff was 28·4 months (IQR 19·7–34·3; 28·3 months [19·4–34·3] in the pembrolizumab group and 28·5 months [20·1–34·3] in the placebo group) and at the third interim analysis was 38·5 months (29·8–44·4; 38·4 months [29·5–44·4] in the pembrolizumab group and 38·6 months [30·2–44·4] in the placebo group). 350 patients in the pembrolizumab group and 346 in the placebo group received at least one dose of study treatment. The mean duration of treatment exposure was 11·6 months (SD 8·5) in the pembrolizumab and 10·0 months (8·0) in the placebo group. 24 (7%) of 350 patients in the pembrolizumab group and 17 (5%) of 346 in the placebo group completed all 2 years of treatment, and 257 (73%) and 286 (83%) patients, respectively, discontinued treatment. At the third interim analysis, the mean duration of treatment exposure was 12·8 months (SD 10·0) in the pembrolizumab and 10·8 months (9·4) in the placebo group. 286 (82%) of 350 in the pembrolizumab group and 304 (88%) of 346 in the placebo group had discontinued treatment, primarily due to disease progression (figure 1).

The study met the prespecified criteria for superiority of pembrolizumab versus placebo for progression-free survival at the second interim analysis. The median progression-free survival was 10·0 months (95% CI 8·6–11·7) in the pembrolizumab group versus 8·1 months (7·0–8·5) in the placebo group (HR 0·72, 95% CI 0·60–0·87; p=0·0002 [superiority boundary, 0·0013]; appendix p 13). At 24 months, an estimated 25·1% (95% CI 20·1–30·5) of patients in the pembrolizumab group versus 14·2% (10·1–19·1) in the placebo group were alive. In the 298 patients in the pembrolizumab group and 296 patients in the placebo group with tumours with a PD-L1 CPS of 1 or more, progression-free survival was also longer with pembrolizumab (median 10·8 months [8·5–12·5] *vs* 7·2 months [6·8–8·4]; HR 0·70 [0·58–0·85]), with 24-month progression-free survival rates of 27·0% (21·5–32·8) versus 13·3% (9·0–18·5; appendix p 13). Progression-free survival did not differ between the 52 patients in each group with tumours with a PD-L1 CPS of less than 1 (median 9·5 months [8·3–11·3] *vs* 9·6 months [7·9–13·0]; HR 1·17 [0·73–1·89]). Consistent with the second interim analysis, progression-free survival was longer in the pembrolizumab versus the placebo group at the third interim analysis, overall (median 10·0 months [8·6–12·2] *vs* 8·1 months [7·1–8·6]; HR 0·73 [0·61–0·87]; figure 2A) and in the subgroup of patients with tumours with a PD-L1 CPS of 1 or more (10·9 months [8·5–12·5] *vs*

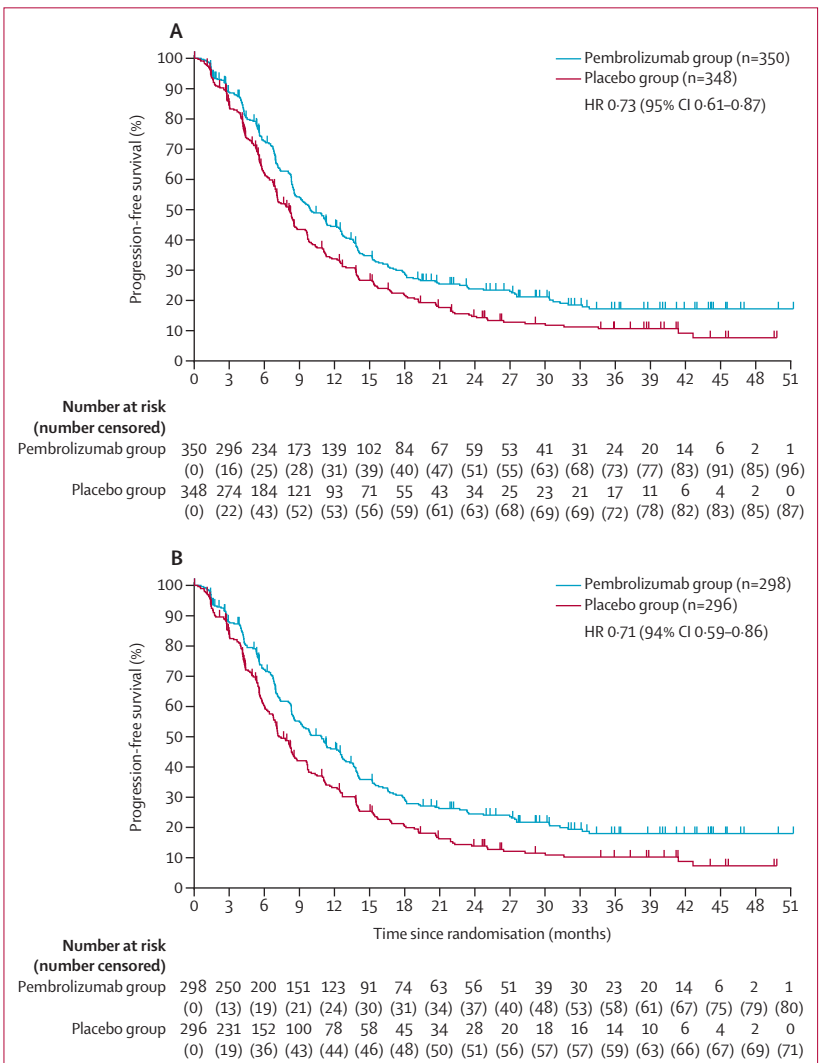

Figure 2: **Progression-free survival in the intention-to-treat population at the third interim analysis**
Kaplan-Meier estimates of progression-free survival in all patients (A) and in the subgroup of patients with tumours with a PD-L1 combined positive score of 1 or more (B). The HR for progression or death is provided on each graph. Tick marks on the Kaplan-Meier curves represent data censored at the time of last imaging assessment. HR=hazard ratio.

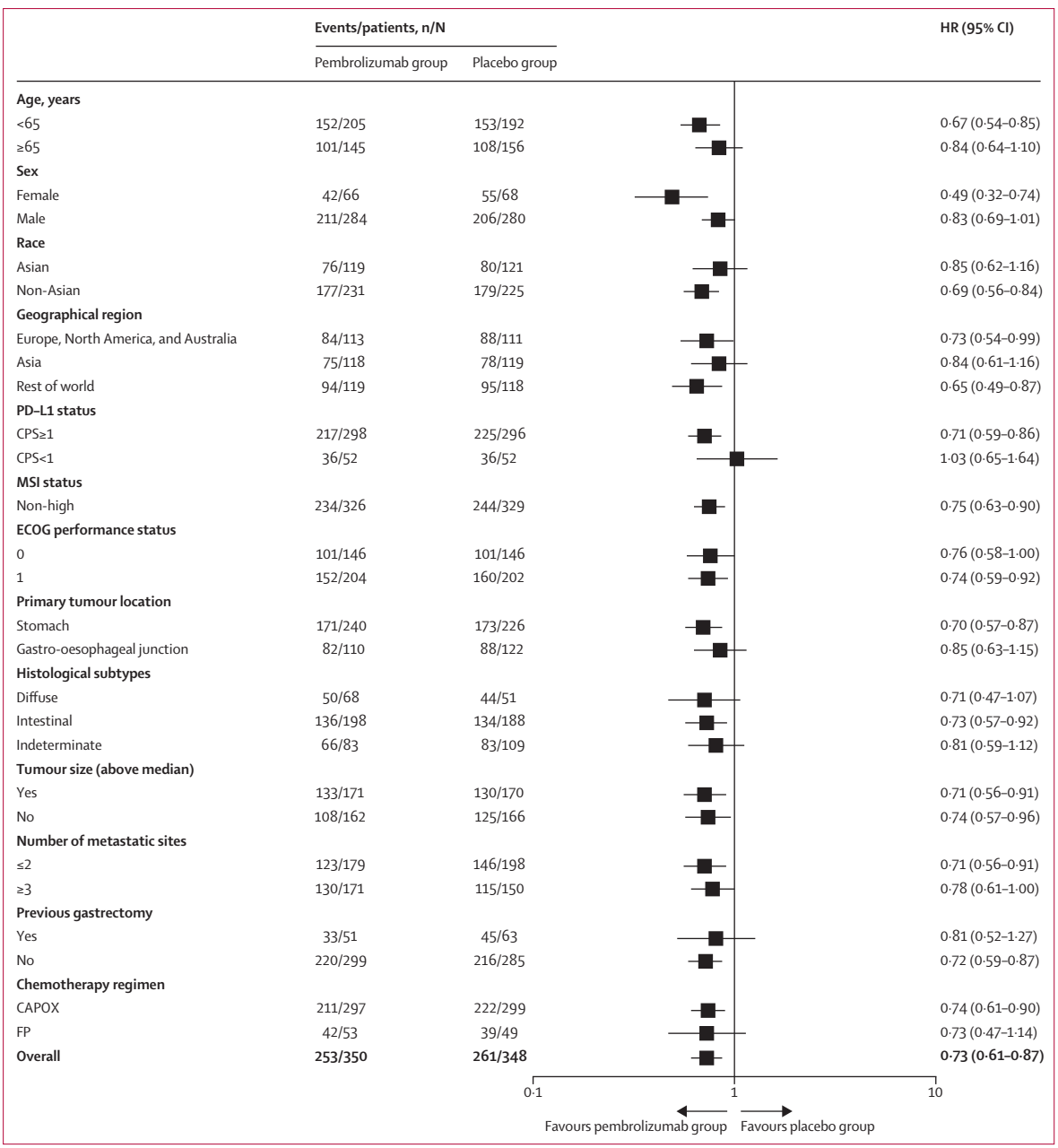

Figure 3: **Forest plot analysis of progression-free survival in prespecified subgroups in the intention-to-treat population at the third interim analysis**
For subgroup variables with two levels, if one level of the subgroup variable had fewer than 20 patients, that subgroup was not included. CAPOX=capecitabine plus oxaliplatin. CPS=combined positive score. ECOG=Eastern Cooperative Oncology Group. FP=fluorouracil plus cisplatin. HR=hazard ratio. MSI=microsatellite instability.

7·3 months [6·8–8·5]; HR 0·71 [0·59–0·86]; figure 2B), but did not differ in the population with a PD-L1 CPS of less than 1 (median 9·5 months [8·3–12·6] *vs* 9·5 months [7·9–13·0]; HR 1·03 [0·65–1·64]). Progression-free survival was consistently improved with pembrolizumab versus placebo irrespective of disease burden, number of metastatic sites, or patient performance status, with the exception of patients with tumours with a PD-L1 CPS of less than 1 (figure 3; appendix p 14). Four patients in the pembrolizumab group and three patients in the placebo group had curative surgery. In sensitivity analyses of progression-free survival with these patients censored at last disease assessment before the time of curative surgery, median progression-free survival was 9·9 months (8·5–11·4) in the pembrolizumab group and 8·1 months (7·0–8·5) in the placebo group (HR 0·73 [0·61–0·87]).

At the second interim analysis, 415 (59%) patients had died, 202 (58%) in the pembrolizumab group and 213 (61%) in the placebo group. Median overall survival was 20·0 months (95% CI 17·8–23·2) in the

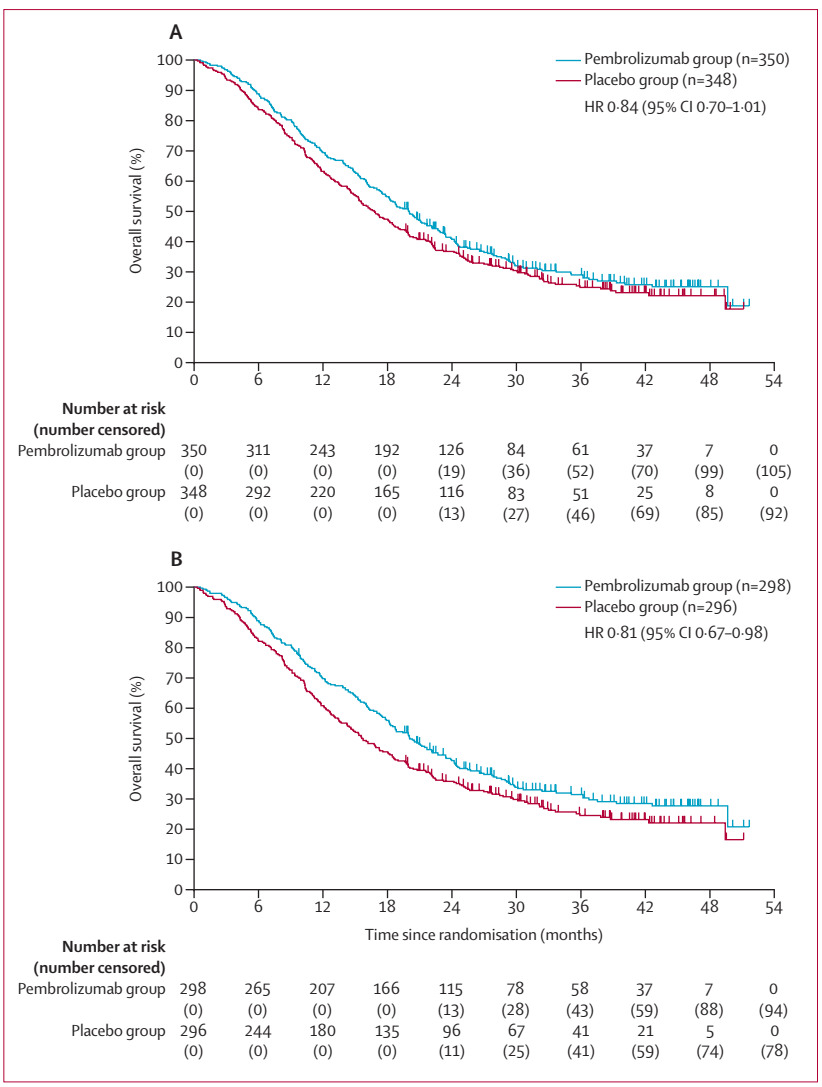

Figure 4: **Overall survival in patients in the intention-to-treat population at the third interim analysis**
Kaplan-Meier estimates of overall survival in all patients (A) and in the subgroup of patients with tumours with a PD-L1 combined positive score of 1 or more (B). The HR for death is provided on each graph. Tick marks represent data censored at the time of last imaging assessment. HR=hazard ratio.

	Pembrolizumab group (N=350)		Placebo group (N=346)	
	Any	Grade ≥3	Any	Grade ≥3
Any adverse event	347 (99%)	248 (71%)	346 (100%)	225 (65%)
Any treatment-related adverse event*	341 (97%)	204 (58%)	334 (97%)	176 (51%)
Serious	88 (25%)	76 (22%)	79 (23%)	66 (19%)
Led to death	4 (1%)	4 (1%)	3 (1%)	3 (1%)
Led to discontinuation of any drug	124 (35%)	59 (17%)	108 (31%)	44 (13%)
Any adverse event of interest†	132 (38%)	36 (10%)	83 (24%)	12 (3%)

(Table 2 continues on next page)

pembrolizumab group versus 16·9 months (15·0–19·8) in the placebo group (HR 0·87 [95% CI 0·72–1·06]; p=0·084), but the significance boundary was not reached. Median overall survival was numerically longer in the pembrolizumab group for the subgroup of patients with tumours with a PD-L1 CPS of 1 or more (20·5 months [18·2–24·3] *vs* 15·6 months [13·5–18·6]; HR 0·79 [0·64–0·98]; appendix p 15), but not in patients with tumours with a PD-L1 CPS of less than 1 (median 16·1 months [13·9–20·8] *vs* 22·3 months [16·6–30·1]; HR 1·61 [0·98–2·64]). At the third interim analysis, median overall survival was 20·0 months (17·8–22·1) in the pembrolizumab group versus 16·8 months (15·0–18·7) in the placebo group (HR 0·84 [0·70–1·01]; figure 4A), and 20·0 months (17·9–22·7) versus 15·7 months (13·5–18·5; HR 0·81 [0·67–0·98]) in the population with a PD-L1 CPS of 1 or more (figure 4B). As these data are still evolving, the independent data monitoring committee recommended that the study continue without changes to the prespecified final analysis of overall survival. Subsequent therapy might affect the overall survival assessment. In this study, 135 (39%) patients in the pembrolizumab group and 164 (47%) in the placebo group received second-line treatment; 55 (16%) patients in the pembrolizumab group and 64 (18%) in the placebo group received a HER2 inhibitor including trastuzumab or trastuzumab deruxtecan, 48 (14%) and 64 (18%), respectively, received a VEGF or VEGFR inhibitor, and 20 (6%) and 21 (6%), respectively, received subsequent anti-PD-1 or anti-PD-L1 therapy (appendix p 8).

The objective response rate (complete or partial response) was 72·6% (95% CI 67·6–77·2; 254 of 350) in the pembrolizumab group versus 59·8% (54·4–65·0; 208 of 348) in the placebo group, with complete responses in 49 (14%) patients in the pembrolizumab group versus 38 (11%) in the placebo group at the second interim analysis, accompanied by changes from baseline in target lesion size (appendix p 16). Among patients with a complete or partial response at the second interim analysis, the median duration of response was 11·2 months (95% CI 9·8–13·2) in the pembrolizumab group versus 9·0 months (7·1–11·2) in the placebo group (appendix p 9). An estimated 31% (95% CI 24·5–37·6) versus 19% (12·8–25·4) of patients with a treatment response had a response duration of at least 24 months. The response rate was consistent at the third interim analysis (72·6% [95% CI 67·6–77·2; 254 of 350] in the pembrolizumab group *vs* 60·1% [54·7–65·2; 209 of 348] in the placebo group), with complete responses in 58 (17%) in the pembrolizumab group versus 39 (11%) in the placebo group. The median duration of response was 11·3 months (9·8–12·7) versus 9·5 months (7·2–11·2), respectively (appendix p 9).

At the second interim analysis, adverse events of any cause occurred in 347 (99%) of 350 patients in the pembrolizumab group and 346 (100%) of 346 patients in

the placebo group (table 2), most commonly diarrhoea (183 [52%] in the pembrolizumab group *vs* 160 [46%] in the placebo group), nausea (168 [48%] *vs* 167 [48%]), anaemia (157 [45%] *vs* 159 [46%]), and vomiting (113 [32%] *vs* 99 [29%]). Events of grade 3 or worse occurred in 248 (71%) patients in the pembrolizumab group and 225 (65%) in the placebo group, most commonly anaemia (44 [13%] *vs* 35 [10%]) and diarrhoea (34 [10%] *vs* 29 [8%]). Treatment-related adverse events occurred in 341 (97%) patients in the pembrolizumab group versus 334 (97%) patients in the placebo group, with grade 3 or worse treatment-related adverse events occurring in 204 (58%) versus 176 (51%) patients. Treatment-related adverse events that led to death occurred in four (1%) patients in the pembrolizumab group (due to hepatitis, sepsis, cerebral infarction, and pneumonitis in one [<1%] patient each), and in three (1%) patients in the placebo group (due to myocarditis, cholangitis, and pulmonary embolism in one [<1%] patient each). At the third interim analysis, the incidence of any cause and treatment-related events were similar to that at the second interim analysis (appendix p 10). Adverse events of interest and infusion reactions occurred in 132 (38%) patients in the pembrolizumab group and 83 (24%) patients in the placebo group (table 2). The most common were infusion reactions, hypothyroidism, pneumonitis, and colitis. Grade 3 or worse immune-mediated adverse events occurred in 36 (10%) patients in the pembrolizumab group versus 12 (3%) patients in the placebo group, most commonly colitis (nine [3%] *vs* seven [2%]). Adverse events of interest that led to death occurred in three (1%) patients in the pembrolizumab group versus one (<1%) patient in the placebo group. Few additional immune-mediated adverse events occurred at the third interim analysis (appendix p 10).

Discussion

Data from these interim analyses of the phase 3 randomised KEYNOTE-811 study showed that first-line pembrolizumab was superior to placebo with respect to progression-free survival in combination with trastuzumab and chemotherapy in patients with HER2-positive metastatic gastric and gastro-oesophageal junction cancer. The progression-free survival curves separated early during treatment, with a preserved separation throughout the evaluation period. In the prespecified subgroup analysis, the progression-free survival benefit with pembrolizumab was significant in patients with tumours with a PD-L1 CPS of 1 or more (85% of all patients), whereas there was no benefit in the small subgroup of patients with tumours with a PD-L1 CPS of less than 1 (15% of all patients). Based on these findings, the US Food and Drug Administration approval of pembrolizumab plus trastuzumab and fluoropyrimidine and platinum-containing chemotherapy for first-line treatment of locally advanced or metastatic HER2-positive

	Pembrolizumab group (N=350)		Placebo group (N=346)	
	Any	Grade ≥3	Any	Grade ≥3
(Continued from previous page)				
Treatment-related adverse events				
Diarrhoea	165 (47%)	31 (9%)	145 (42%)	27 (8%)
Nausea	154 (44%)	14 (4%)	152 (44%)	15 (4%)
Anaemia	109 (31%)	21 (6%)	113 (33%)	20 (6%)
Neutrophil count decreased	92 (26%)	28 (8%)	83 (24%)	30 (9%)
Decreased appetite	91 (26%)	11 (3%)	91 (26%)	11 (3%)
Platelet count decreased	89 (25%)	22 (6%)	93 (27%)	23 (7%)
Vomiting	88 (25%)	14 (4%)	86 (25%)	10 (3%)
Peripheral sensory neuropathy	84 (24%)	13 (4%)	73 (21%)	7 (2%)
Plantar-palmar erythrodysaesthesia syndrome	78 (22%)	5 (1%)	72 (21%)	5 (1%)
Fatigue	69 (20%)	12 (3%)	57 (16%)	8 (2%)
Aspartate aminotransferase increased	66 (19%)	6 (2%)	50 (14%)	1 (<1%)
Peripheral neuropathy	60 (17%)	8 (2%)	63 (18%)	9 (3%)
Neutropenia	59 (17%)	22 (6%)	54 (16%)	16 (5%)
White blood cell count decreased	53 (15%)	3 (1%)	41 (12%)	6 (2%)
Alanine aminotransferase increased	51 (15%)	3 (1%)	41 (12%)	1 (<1%)
Weight decreased	42 (12%)	5 (1%)	24 (7%)	2 (1%)
Thrombocytopenia	40 (11%)	11 (3%)	44 (13%)	6 (2%)
Asthenia	39 (11%)	7 (2%)	50 (14%)	9 (3%)
Blood bilirubin increased	39 (11%)	3 (1%)	27 (8%)	2 (1%)
Stomatitis	36 (10%)	4 (1%)	31 (9%)	6 (2%)
Adverse events of interest†				
Infusion reactions	56 (16%)	9 (3%)	45 (13%)	2 (1%)
Hypothyroidism	37 (11%)	1 (<1%)	15 (4%)	0
Pneumonitis	21 (6%)	6 (2%)	5 (1%)	0
Colitis	17 (5%)	9 (3%)	10 (3%)	7 (2%)
Hyperthyroidism	14 (4%)	0	11 (3%)	0
Adrenal insufficiency	4 (1%)	1 (<1%)	0	0
Guillain-Barre syndrome	0	0	1 (<1%)	0
Hepatitis	2 (1%)	2 (1%)	4 (1%)	2 (1%)
Hypophysitis	4 (1%)	2 (1%)	0	0
Myocarditis	0	0	1 (<1%)	1 (<1%)
Myositis	1 (<1%)	1 (<1%)	0	0
Nephritis	4 (1%)	1 (<1%)	0	0
Pancreatitis	0	0	1 (<1%)	0
Severe skin reactions	3 (1%)	3 (1%)	0	0
Thyroiditis	4 (1%)	0	0	0
Type 1 diabetes	1 (<1%)	1 (<1%)	0	0
Uveitis	1 (<1%)	0	1 (<1%)	0
Vasculitis	4 (1%)	0	1 (<1%)	0

Data are number (%) of patients in the as-treated population (ie, all patients who were randomly assigned and received at least one dose of study treatment) with at least one occurrence of the listed event. Adverse event data for the third interim analysis are provided in the appendix (p 10). *Treatment-related adverse events were attributed to study treatment by an investigator; reported are treatment-related adverse events that occurred in at least 10% of patients in any group. †Adverse events of interest were based on a list of terms specified by the funder, regardless of attribution to any study treatment by an investigator. Grade 5 adverse events of interest in the pembrolizumab group were hepatitis (n=1) and pneumonitis (n=2), which occurred in three (1%) patients; grade 5 adverse events of interest in the placebo group were myocarditis (n=1), which occurred in one (<1%) patient.

Table 2: **Summary of adverse events in all treated patients at the second interim analysis**

gastric or gastro-oesophageal junction adenocarcinoma will be revised and limited to patients with tumours with a PD-L1 CPS of 1 or more.[17]

Although treatment-related adverse events were observed in all patients, the frequency of chemotherapy-related and immunotherapy-related grade 3 or worse adverse events was similar to that reported for combinations of trastuzumab or pembrolizumab with chemotherapy.[18] No new safety concerns were observed. The improvement in progression-free survival observed at the second interim analysis was maintained at the third interim analysis. At the time of the third interim analysis there was no significant improvement in overall survival. The overall survival analysis is not yet definitive and that is the major limitation of this publication. Only 415 (75%) of 551 death events occurred and the independent data monitoring committee recommended continued masking of overall survival data until the prespecified final analysis.

Over the past decade, studies attempting to improve treatment options in patients with HER2-positive gastric cancer using dual HER2-targeted therapy such as pertuzumab or lapatinib in combination with trastuzumab have not met their primary endpoints.[19] The progression-free survival benefit observed with pembrolizumab in combination with trastuzumab and chemotherapy in the current study was consistent across several prespecified subgroups. Notably, only six patients in the pembrolizumab group and two in the placebo group had tumours positive for high microsatellite instability, a molecularly defined subset with a greater likelihood of response to anti-PD-1 therapy. Given the consistent progression-free survival observed overall (HR 0·73) and in the non-high microsatellite instability subgroup (HR 0·75) at the third interim analysis, the difference in progression-free survival observed in the current study cannot be attributed to having high microsatellite instability. In response to trastuzumab, tumours have been known to upregulate PD-L1 expression to dampen the antibody-dependent cell-mediated cytotoxicity of trastuzumab.[20] Analysis of samples collected from patients enrolled in KEYNOTE-811 before and during therapy will help to establish whether pembrolizumab increases immune-cell engagement or enhances the effects of trastuzumab or chemotherapy through other mechanisms.

Oxaliplatin has a more favourable adverse event profile than cisplatin and has replaced cisplatin as the preferred chemotherapy backbone in combination with fluorouracil.[21–23] Each platinum agent might have unique effects on dendritic-cell stimulation and trafficking.[24] Interpretation of the chemotherapy subgroup analysis is complicated by the wide confidence interval for the fluorouracil plus cisplatin subgroup, reflecting that only 15% of patients received cisplatin-based therapy in this study. Oxaliplatin was the preferred choice by most investigators, and it is unknown whether the choice of the platinum agent affected the outcomes of the current study. Numerically, the efficacy outcomes observed in the placebo group of KEYNOTE-811 (who also received trastuzumab, a fluoropyrimidine, and mainly oxaliplatin) appear to be better than the historical outcomes observed with trastuzumab, cisplatin, and fluorouracil in the phase 3 ToGA study[5] with median progression-free survival of 8·1 months in KEYNOTE-811 versus 6·7 months in ToGA, in the respective ITT populations. The difference in the baseline study population and proportion of HER2 IHC 3+ tumours in KEYNOTE-811 (78%) and ToGA (45%) probably contributes to the favourable outcomes of the control group in the context of the historical data, as patients with IHC 3+ tumours typically have a greater degree of clinical benefit in response to HER2-targeted agents than those with lower levels of HER2 expression. These factors were controlled by randomisation and the proportion of patients with IHC 3+ tumours were balanced between the treatment groups in KEYNOTE 811. Importantly, the proportion of patients with IHC 3+ tumours is similar to that reported in the published literature.[25] An additional limitation could be that although a majority of the patient population had tumours with a PD-L1 CPS of 1 or more and this was a prespecified subgroup, PD-L1 expression was not included as a primary endpoint analysis. The small proportion of the population with a PD-L1 CPS of less than 1 limited full assessment of this group. The key limitation of this interim analysis is the lack of a definitive overall survival outcome, as the study was continued to final analysis. Ultimately, the effect on overall survival is a consideration when determining the best first-line therapy options. Subsequent anticancer therapy might also affect the overall survival analysis. The proportion of patients who received subsequent anticancer therapy in this study (39% in the pembrolizumab group *vs* 47% in the placebo group) is similar to that of contemporaneous global phase 3 studies.[10,23] Although 80% of the study population were male, this proportion is reflective of this type of cancer.

These data represent a step forward for biomarker-driven studies for gastro-oesophageal cancers. The addition of pembrolizumab to first-line trastuzumab and chemotherapy led to meaningfully longer progression-free survival and improvement in tumour response, particularly in the subgroup of patients with HER2-positive gastro-oesophageal cancer who had PD-L1-positive tumours. These data resulted in approval of the combination of pembrolizumab and trastuzumab plus chemotherapy as a first-line therapy in patients with advanced unresectable or metastatic gastric or gastro-oesophageal cancer with HER2 overexpression in the USA, Europe, and Brazil. The final overall survival analysis is needed to establish the long-term therapeutic effect of this regimen.

Contributors

YYJ, HCC, KS, EVC, JT, C-SS, and SYR contributed to the conception, design, and planning of the study. YYJ, AK, YB, SL, PY, LSW, LS, YO, MB, HCC, KS, S-KQ, KL, and SYR contributed to the acquisition of data.

YYJ, JPM, JT, and KL contributed to the analysis of data. YYJ, KL, C-SS, and PB verified the data. YYJ, AK, JX, SL, JPM, PY, LS, MB, HCC, KS, S-KQ, EVC, JT, KL, C-SS, PB, and SYR contributed to the interpretation of results. All authors had access to the data, were involved in reviewing and editing the manuscript, approved the submitted draft and accepted responsibility to submit the manuscript for publication, and vouch for the accuracy of the data reported. All authors contributed to the drafting or critical review and revision of the manuscript, and approved the final version to be submitted.

Declaration of interests

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Data sharing

Merck Sharp & Dohme, a subsidiary of Merck & Co, Rahway, NJ, USA (MSD) is committed to providing qualified scientific researchers access to anonymised data and clinical study reports from the company's clinical trials for the purpose of conducting legitimate scientific research. MSD is also obligated to protect the rights and privacy of trial participants and, as such, has a procedure in place for evaluating and fulfilling requests for sharing company clinical trial data with qualified external scientific researchers. The MSD data sharing website (http://engagezone.msd.com/ds_documentation.php) outlines the process and requirements for submitting a data request. Applications will be promptly assessed for completeness and policy compliance. Feasible requests will be reviewed by a committee of MSD subject matter experts to assess the scientific validity of the request and the qualifications of the requestors. In line with data privacy legislation, submitters of approved requests must enter into a standard data-sharing agreement with MSD before data access is granted. Data will be made available for request after product approval in the USA and EU or after product development is discontinued. There are circumstances that might prevent MSD from sharing requested data, including country or region-specific regulations. If the request is declined, it will be communicated to the investigator. Access to genetic or exploratory biomarker data requires a detailed, hypothesis-driven statistical analysis plan that is collaboratively developed by the requestor and MSD subject matter experts; after approval of the statistical analysis plan and execution of a data-sharing agreement, MSD will either perform the proposed analyses and share the results with the requestor or will construct biomarker covariates and add them to a file with clinical data that is uploaded to an analysis portal so that the requestor can perform the proposed analyses.

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